

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0427

Halifax 138 kV
April 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0427. The Interconnection Customer proposes a 87.6 MW solar photovoltaic facility interconnecting at the Halifax 138 kV substation in the Virginia area. The request was received on January 21, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 239 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,102,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Drum - Montpelier 138 kV line: 132.4% loading (432 MVA) following the outage of the Taber Substation - Glenannan 345 kV; DFAX = 5.0%.

N Pole - Dixleta 138 kV #1: 130.7% loading (465 MVA) following the outage of the N Pole - Dixleta 138 kV Line 2; DFAX = 5.5%.

The N Pole - Dixleta 138 kV #2 is overloaded to 128.6% of its 472 MVA emergency rating for the loss of the N Pole - Dixleta 138 kV Circuit 1. The project contributes 5.5% DFAX (4.8 MW) to this overload.

Hughson Substation - Dixleta 345 kV: 127.0% loading (1,673 MVA) following the outage of the Rustic - Dileta 345 kV line; DFAX = 18.4%.

For the loss of Glenannan Substation - Taber Substation 345 kV, the N Pole - High Ridge 138 kV loads to 125.1% of its 799 MVA rating. Project DFAX: 7.8%.

For the loss of Taber - Dixleta 345 kV, the Westport - Wheelabrator Baltimore Refuse 138 kV loads to 123.4% of its 589 MVA rating. Project DFAX: 5.4%.

Rustic - Dixleta 345 kV: 123.2% loading (1,623 MVA) following the outage of the Hughson - Dixleta 345 kV L; DFAX = 16.8%.

For the loss of Taber - Glenannan Substation 345 kV Line, the High Ridge - Jerome 138 kV loads to 119.5% of its 868 MVA rating. Project DFAX: 7.8%.

The FairInd - Sunbright 138 kV Line is overloaded to 118.5% of its 877 MVA emergency rating for the loss of the Taber - Glenannan 345 kV L. The project contributes 6.7% DFAX (5.9 MW) to this overload.

Sunbright - Colesville 138 kV line: 116.4% loading (941 MVA) following the outage of the Glenannan - Taber 345 kV; DFAX = 6.7%.

For the loss of Glenannan - Taber 345 kV L, the Beltsville - Bunker Hill 138 kV L loads to 114.6% of its 812 MVA rating. Project DFAX: 5.3%.

For the loss of Sierra Pacific Lumber 138/1kV Transformer, the Taber - Glenannan 345 kV loads to 114.4% of its 2,382 MVA rating. Project DFAX: 18.4%.

The Dixleta - Drum 138 kV is overloaded to 113.6% of its 818 MVA emergency rating for the loss of the Taber - Glenannan Substation 345 kV Line. The project contributes 5.6% DFAX (4.9 MW) to this overload.

The Nasa - Bunker Hill 138 kV line is overloaded to 113.3% of its 855 MVA emergency rating for the loss of the Taber - Glenannan 345 kV. The project contributes 5.3% DFAX (4.6 MW) to this overload.

For the loss of Sierra Pacific Lumber 138-1 kV TX, the Taber - Dixleta 345 kV loads to 112.8% of its 2,258 MVA rating. Project DFAX: 16.6%.

Aibonito Pds - Shenandoah 138 kV line: 111.9% loading (238 MVA) following the outage of the Bear Substation - Clover 345 kV; DFAX = 5.5%.

The Nauvoo - Birchwood 345 kV is overloaded to 111.4% of its 1,663 MVA emergency rating for the loss of the Fredericksburg - Dunn Siding 345 kV L. The project contributes 14.9% DFAX (13.1 MW) to this overload.

Chancellor Row - Bear Island 345 kV: 111.3% loading (1,376 MVA) following the outage of the Leavitt - Birchwood 345 kV; DFAX = 18.6%.

For the loss of Bixby Hill - Tombstone 500 kV line, the Bixby Hill - Sharyland 345 kV Line loads to 109.7% of its 2,462 MVA rating. Project DFAX: 6.7%.

Tombstone - Sharyland 345 kV: 109.7% loading (2,508 MVA) following the outage of the Tombstone - Bixby Hill 500 kV; DFAX = 6.6%.

For the loss of Rustic - Dixleta 345 kV, the Hughson - Rustic 345 kV loads to 109.5% of its 1,540 MVA rating. Project DFAX: 12.6%.

Riva Rd - Sierra Pacific Lumber 345 kV: 107.8% loading (1,220 MVA) following the outage of the Taber - Glenannan 345 kV L; DFAX = 6.5%.

For the loss of Louisa Generation Facility - Carrabelle 345 kV, the Louisa Generation Facility Substation - N Anna 345 kV loads to 107.3% of its 2,070 MVA rating. Project DFAX: 12.7%.

For the loss of Clover Substation - Bear 345 kV, the Shenandoah - Walworth 138 kV loads to 106.7% of its 231 MVA rating. Project DFAX: 5.5%.

For the loss of Bear - Clover 345 kV L, the Bear - Mecklenburg Power Sta 345 kV L loads to 106.3% of its 1,927 MVA rating. Project DFAX: 27.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Drum - Montpelier 138 kV line	Rebuild line with higher-capacity conductor	\$8,671,000	3% (\$296,000)
N Pole - Dixleta 138 kV #1	Rebuild line with higher-capacity conductor	\$1,002,000	2% (\$19,000)
Dixleta - N Pole 138 kV #2	Rebuild line with higher-capacity conductor	\$1,256,000	2% (\$24,000)
Hughson Substation - Dixleta 345 kV	Rebuild line with higher-capacity conductor	\$31,339,000	2% (\$484,000)
North Pole - High Ridge 138 kV Line	Rebuild line with higher-capacity conductor	\$2,218,000	2% (\$38,000)
Westport - Wheelabrator Baltimore Refuse 138 kV Line	Rebuild line with higher-capacity conductor	\$400,000	2% (\$9,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rustc - Dixleta 345 kV	Rebuild line with higher-capacity conductor	\$76,157,000	2% (\$1,172,000)
High Ridge - Jerome 138 kV	Rebuild line with higher-capacity conductor	\$2,464,000	2% (\$42,000)
Fairlnd - Sunbright 138 kV Line	Rebuild line with higher-capacity conductor	\$732,000	2% (\$12,000)
Sunbright - Colesville 138 kV line	Rebuild line with higher-capacity conductor	\$838,000	2% (\$14,000)
Beltsville - Bunker Hill 138 kV L	Rebuild line with higher-capacity conductor	\$949,000	3% (\$32,000)
Taber - Glenannan 345 kV	Rebuild line with higher-capacity conductor	\$36,541,000	2% (\$570,000)
Drum Substation - Dixleta 138 kV line	Rebuild line with higher-capacity conductor	\$1,890,000	2% (\$33,000)
Nasa - Bunker Hill 138 kV line	Rebuild line with higher-capacity conductor	\$2,228,000	3% (\$76,000)
Dixleta - Taber Substation 345 kV line	Rebuild line with higher-capacity conductor	\$21,487,000	2% (\$335,000)
Shenandoah - Aibonito Pds 138 kV	Rebuild line with higher-capacity conductor	\$30,801,000	18% (\$5,568,000)
Nauvoo - Birchwood 345 kV	Rebuild line with higher-capacity conductor	\$82,846,000	2% (\$1,502,000)
Chancellor Row - Bear Island 345 kV	Rebuild line with higher-capacity conductor	\$17,041,000	2% (\$383,000)
Bixby Hill - Sharyland 345 kV Line	Reconductor line	\$22,832,000	2% (\$341,000)
Tombstone - Sharyland 345 kV	Reconductor line	\$24,372,000	2% (\$361,000)
Hughson - Rustic 345 kV	Reconductor line	\$20,128,000	2% (\$305,000)
Riva Rd - Sierra Pacific Lumber 345 kV line	Reconductor line	\$19,481,000	2% (\$331,000)
Louisa Generation Facility Substation - N Anna 345 kV	Reconductor line	\$39,505,000	1% (\$561,000)
Walworth - Shenandoah Substation 138 kV L	Reconductor line	\$9,991,000	18% (\$1,806,000)
Bear - Mecklenburg Power Station 345 kV	Reconductor line	\$45,162,000	4% (\$1,788,000)
	Total allocated to project		\$16,102,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.